

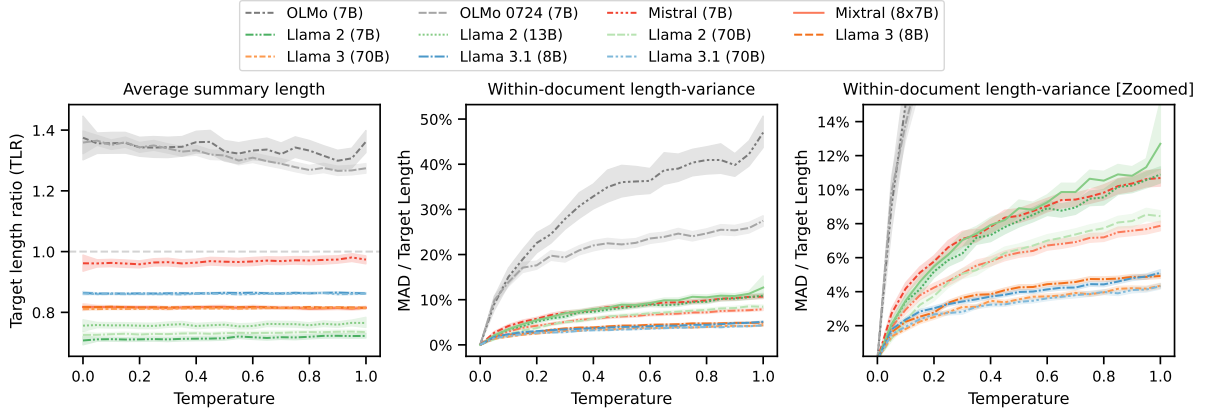
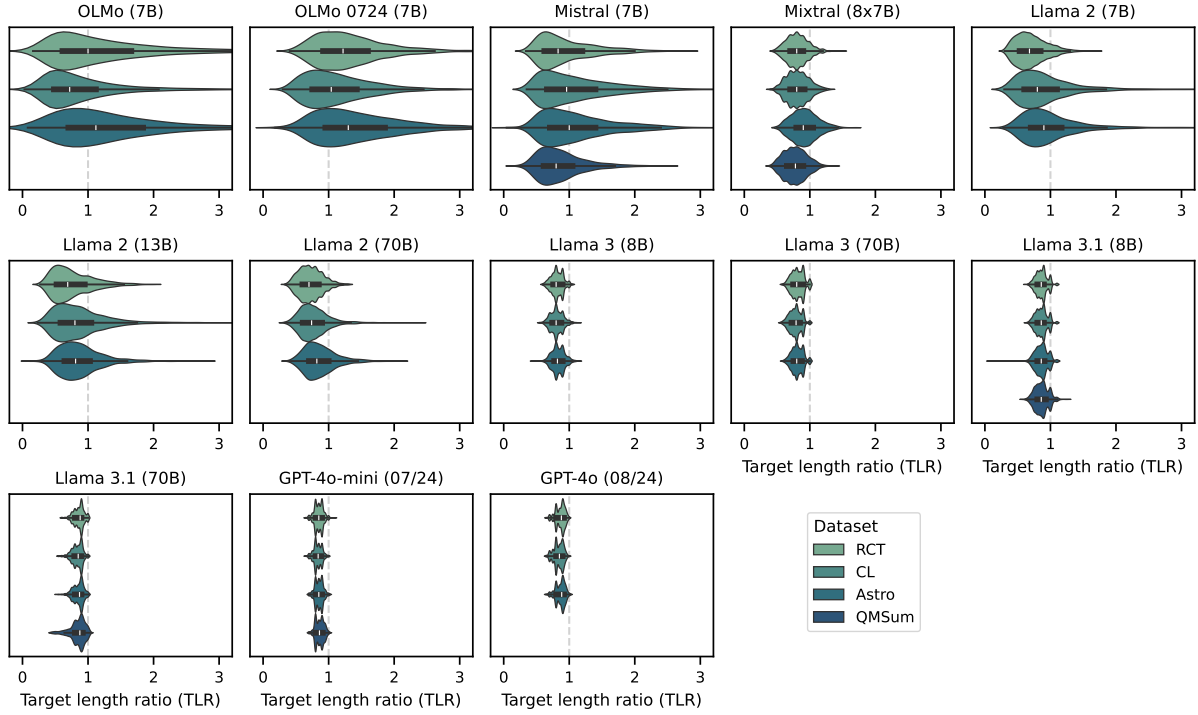


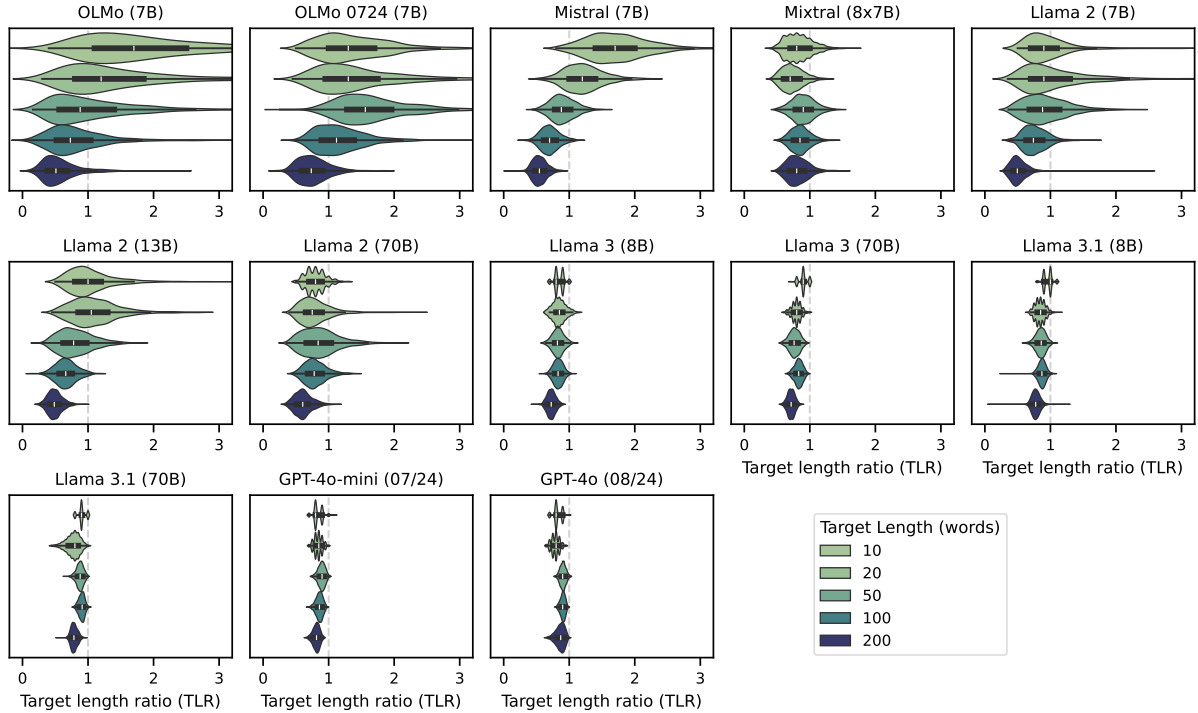
Figure 9: Influence of temperature on generated summary length. **Left:** target length-ratio. **Center:** “within-document length variance” calculated as the mean deviation from the average summary length of 5 summaries for the same document (MAD). MAD is normalized to be comparable across length targets. **Right:** zoomed version.

| Measure                                            | Dataset            | Random      | OLMo          | Mixtral       | Llama <sup>3.1</sup> <sub>8b</sub> | Llama <sup>3.1</sup> <sub>70b</sub> | 4o-mini       | 4o            | Average       |
|----------------------------------------------------|--------------------|-------------|---------------|---------------|------------------------------------|-------------------------------------|---------------|---------------|---------------|
| Consistency of Saliency Estimates                  |                    |             |               |               |                                    |                                     |               |               |               |
| <i>LLM-perceived</i>                               | RCT                | -0.06       | 0.34*         | 0.48**        | 0.29*                              | 0.61**                              | 0.75**        | <b>0.80**</b> | 0.46**        |
|                                                    | Astro              | 0.02        | 0.07          | 0.41*         | 0.56*                              | 0.64**                              | 0.76**        | <b>0.86**</b> | 0.47**        |
|                                                    | CL                 | -0.10       | -0.05         | 0.65**        | 0.29                               | <b>0.73**</b>                       | 0.70**        | 0.57**        | 0.40**        |
|                                                    | QMsum              | -0.07       | 0.42**        | 0.61*         | 0.36                               | <b>0.87**</b>                       | 0.72**        | 0.80**        | 0.53**        |
|                                                    | Average            | -0.05       | 0.20*         | 0.54**        | 0.37*                              | 0.71**                              | 0.73**        | <b>0.76**</b> | 0.47**        |
| <i>LLM-observed</i>                                | RCT                | 0.94**      | 0.98**        | <b>0.99**</b> | <b>0.99**</b>                      | <b>0.99**</b>                       | <b>0.99**</b> | <b>0.99**</b> | <b>0.99**</b> |
|                                                    | Astro              | 0.91**      | 0.99**        | <b>1.00**</b> | 0.97**                             | 0.99**                              | 0.97**        | 0.97**        | 0.97**        |
|                                                    | CL                 | 0.96**      | <b>0.99**</b> | <b>0.99**</b> | 0.96**                             | <b>0.99**</b>                       | <b>0.99**</b> | 0.97**        | 0.98**        |
|                                                    | QMsum <sup>†</sup> | 0.87**      | —             | 0.99**        | 0.99**                             | <b>1.00**</b>                       | 0.99**        | —             | 0.97**        |
|                                                    | Average            | 0.92**      | <b>0.99**</b> | <b>0.99**</b> | 0.98**                             | <b>0.99**</b>                       | 0.98**        | 0.98**        | 0.97**        |
| Correlation of Saliency Estimates                  |                    |             |               |               |                                    |                                     |               |               |               |
| <i>LLM-perceived</i><br>vs.<br><i>LLM-observed</i> | RCT                | -0.06       | 0.10          | 0.25          | 0.25                               | 0.37**                              | 0.41**        | <b>0.51**</b> | 0.28*         |
|                                                    | Astro              | 0.11        | 0.09          | 0.31          | 0.56**                             | 0.50*                               | <b>0.65**</b> | 0.58*         | 0.40**        |
|                                                    | CL                 | -0.08       | 0.16          | 0.44          | 0.47*                              | 0.38                                | <b>0.58*</b>  | 0.41          | 0.34          |
|                                                    | QMsum <sup>†</sup> | 0.11        | —             | 0.46*         | 0.16                               | <b>0.63*</b>                        | 0.60*         | —             | 0.39*         |
|                                                    | Average            | 0.03        | 0.12          | 0.37*         | 0.36**                             | 0.47*                               | <b>0.56**</b> | 0.50*         | 0.35*         |
| Correlation of Model and Human Saliency            |                    |             |               |               |                                    |                                     |               |               |               |
| <i>LLM-perceived</i><br>vs. <i>Human</i>           | RCT                | -0.03       | 0.22          | 0.38**        | 0.34*                              | 0.49**                              | 0.48*         | <b>0.56**</b> | 0.35**        |
|                                                    | Astro              | 0.07        | 0.12          | 0.30**        | 0.31*                              | 0.27                                | <b>0.45**</b> | 0.44*         | 0.28**        |
|                                                    | CL                 | 0.06        | -0.03         | 0.41*         | 0.22                               | <b>0.48*</b>                        | 0.44*         | 0.46**        | 0.29**        |
|                                                    | QMsum              | 0.14        | 0.34          | 0.54*         | 0.36*                              | 0.62*                               | <b>0.67**</b> | <b>0.67**</b> | 0.48*         |
|                                                    | Average            | 0.07        | 0.16          | 0.41*         | 0.31*                              | 0.46**                              | 0.51**        | <b>0.53**</b> | 0.35**        |
| <i>LLM-observed</i> vs.<br><i>Human</i>            | RCT                | 0.31        | 0.28          | 0.27          | 0.25                               | 0.25                                | <b>0.34</b>   | 0.24          | 0.27          |
|                                                    | Astro              | 0.11        | 0.25*         | 0.27*         | 0.29*                              | <b>0.31</b>                         | 0.26          | 0.25*         | 0.25*         |
|                                                    | CL                 | <b>0.30</b> | 0.23          | 0.23          | 0.24                               | 0.26                                | 0.25          | 0.24          | 0.25          |
|                                                    | QMsum <sup>†</sup> | 0.16        | —             | 0.53*         | 0.58**                             | <b>0.59**</b>                       | 0.51**        | —             | 0.48*         |
|                                                    | Average            | 0.20        | 0.25          | 0.33*         | 0.35*                              | <b>0.36*</b>                        | 0.34*         | 0.25          | 0.30*         |

Table 6: Spearman rank correlations between saliency estimates, split by dataset. Significance: \* ( $p < 0.05$ ) and \*\* ( $p < 0.01$ ); row-wise maxima are bolded. <sup>†</sup>Results for QMSum not available due to limited context window (OLMo) and budget constraints (GPT-4o).



(a) Distribution of target length ratios over all generated summaries stratified by dataset.



(b) Distribution of target length ratios over all generated summaries stratified by target summary length.

Figure 10: Analysis of length-instruction following. The target length ration (TLR) indicates to what extent models match the provided length. A value of 1 indicates perfect length match, while values greater or smaller than 1 indicate over- or under-generation, respectively.

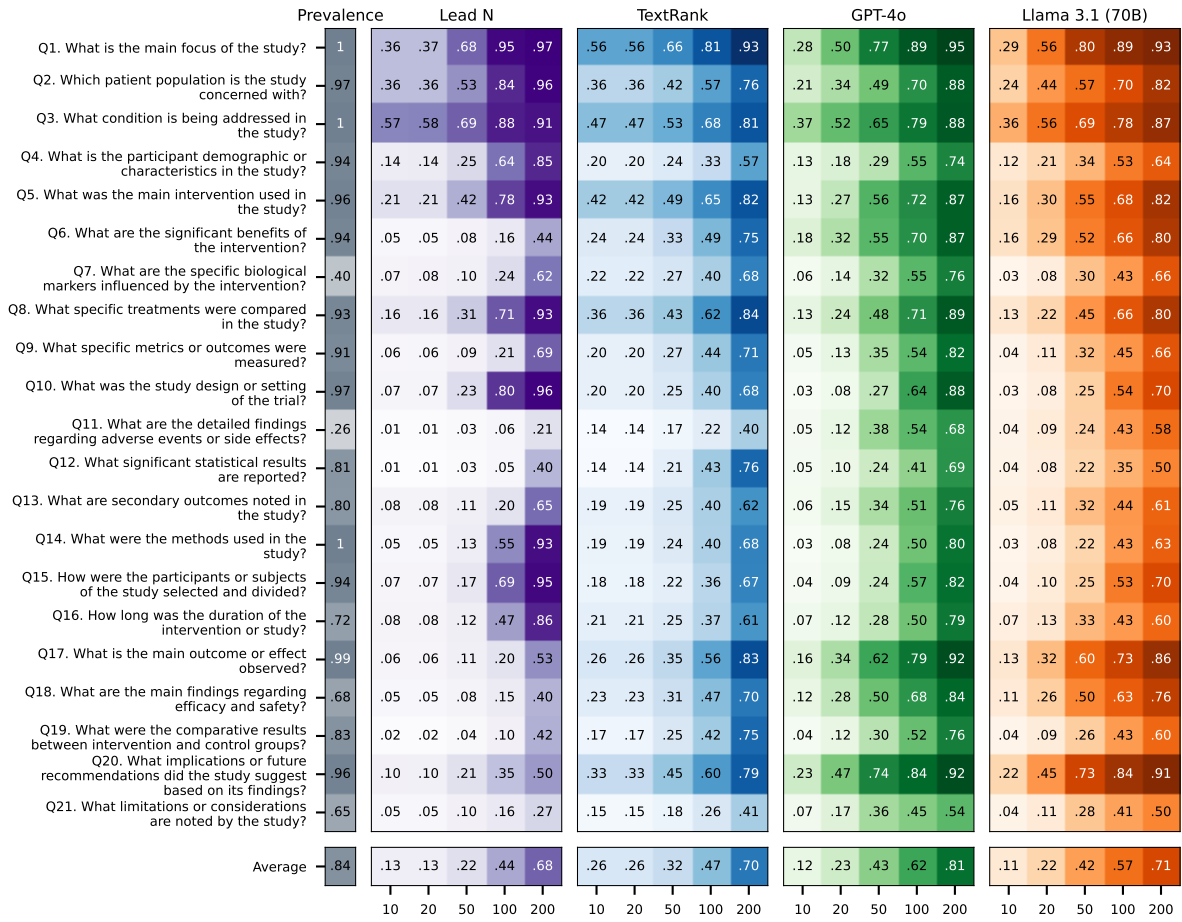


Figure 11: Corpus-level content salience map for *RCT* summaries by four methods.

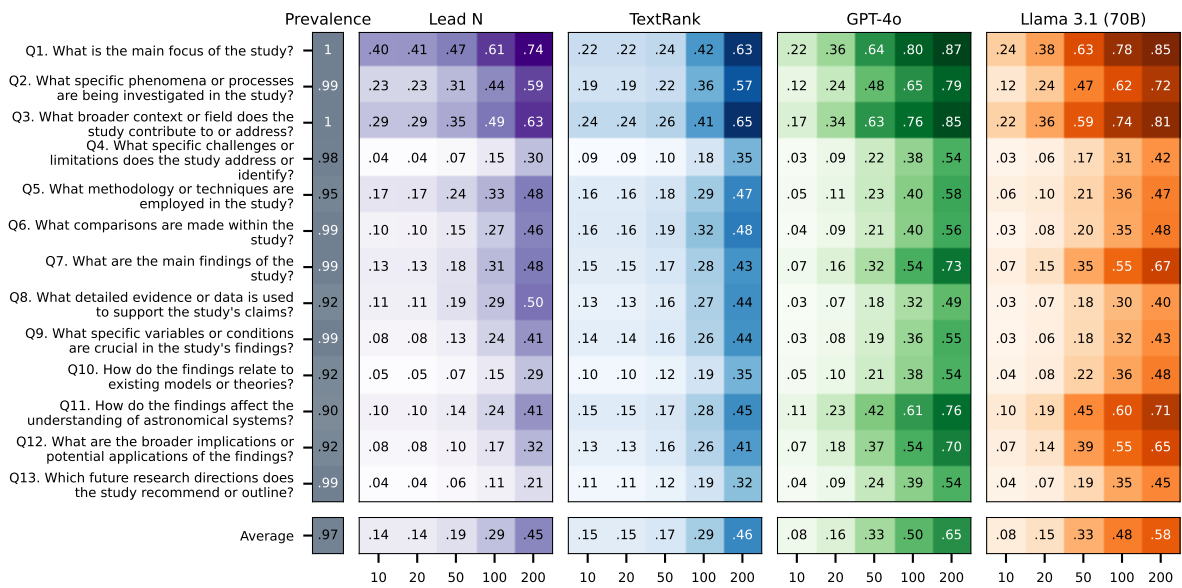


Figure 12: Corpus-level content salience map for *Astro* summaries by four methods.